

Day 5 — Lunar Months and the Festival Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the 12 lunar months in order, explain how each month is named (after the *nakṣatra* of its *pūrṇimā*), and link at least three festivals to their tithi-and-month positions.

Materials

- Board with a 12-row table ready
- A printed list of major festivals across the 12 lunar months (one per pair)
- Notebooks

Vocabulary introduced today

- *māsa* (IAST: māsa; lit. “month”) — a lunar month, ~29.53 days.
- *nakṣatra* (IAST: nakṣatra; lit. “lunar mansion”) — one of 27 segments of the ecliptic, each ~13°20’ wide. (Full coverage in a later module — today’s mention is just for naming.)
- *adhika māsa* (IAST: adhika-māsa; lit. “extra month”) — an intercalary 13th month inserted approximately every 32.5 months.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- Quick recall using yesterday’s tithi calendar: “*What month did we build the calendar for?*” (May, June, etc.) “*In the Indian lunar calendar, what month is that?*” (Answer: it depends — let’s find out.)

Core (20 min)

1. **The 12 lunar months.** Write the list on the board:

#	Lunar month (IAST)	Roughly corresponds to	Named after
1	<i>Caitra</i>	March–April	nakṣatra <i>Citrā</i>
2	<i>Vaiśākha</i>	April–May	<i>Viśākhā</i>
3	<i>Jyēṣṭha</i>	May–June	<i>Jyēṣṭhā</i>
4	<i>Āṣāḍha</i>	June–July	<i>Pūrva- / Uttara-āṣāḍhā</i>
5	<i>Śrāvaṇa</i>	July–August	<i>Śravaṇa</i>
6	<i>Bhādrapada</i>	August–	<i>Pūrva- / Uttara-</i>

#	Lunar month (IAST)	Roughly corresponds to	Named after
		September	<i>bhādrapadā</i>
7	<i>Āśvina</i>	September–October	<i>Aśvinī</i>
8	<i>Kārtika</i>	October–November	<i>Kṛttikā</i>
9	<i>Mārgaśīrṣa</i>	November–December	<i>Mṛgaśīrṣa</i>
10	<i>Pauṣa</i>	December–January	<i>Puṣya</i>
11	<i>Māgha</i>	January–February	<i>Maghā</i>
12	<i>Phālguna</i>	February–March	<i>Phālgunī</i> (Pūrva / Uttara)

2. **Why these names? The naming convention.** A lunar month is named after the *nakṣatra* in which the *pūrṇimā* (full moon) of that month falls. The Moon at full moon is opposite the Sun, so it sits in a specific star background — the *nakṣatra*. The lunar month inherits its name from that *nakṣatra*.

3. **Two month-counting conventions.** This is a real difference and worth noting:

- *Amānta* — the lunar month ends at *amāvasyā*. Common in South and Western India (Maharashtra, Karnataka, Andhra, Tamil Nadu, Gujarat).
- *Pūrṇimānta* — the lunar month ends at *pūrṇimā*. Common in North India (Uttar Pradesh, Bihar, North Indian Hindi belt).

So the same fortnight can be called by different month names under the two conventions! For example, the *kṛṣṇa pakṣa* before Janmāṣṭamī is “*Śrāvaṇa kṛṣṇa pakṣa*” under *pūrṇimānta* and “*Bhādrapada kṛṣṇa pakṣa*” under *amānta*. Both are correct — they are counting from different anchor points.

4. **Linking festivals to tithis — the festival table.** Write 6 examples:

Festival	Lunar month + pakṣa + tithi	What it celebrates / who observes
<i>Ugādi</i> / <i>Gudi</i> / <i>Pādvā</i>	<i>Caitra śukla pratipadā</i>	Lunar new year (in much of South / West India)
<i>Rāma Navamī</i>	<i>Caitra śukla navamī</i>	Birth of Rāma
<i>Akṣaya Tṛtīyā</i>	<i>Vaiśākha śukla tṛtīyā</i>	Auspicious for beginnings
<i>Guru Pūrṇimā</i>	<i>Āṣāḍha pūrṇimā</i>	Honouring teachers
<i>Janmāṣṭamī</i>	<i>Bhādrapada</i> (or <i>Śrāvaṇa</i> , depending) <i>kṛṣṇa aṣṭamī</i>	Birth of Kṛṣṇa

Festival	Lunar month + pakṣa + tithi	What it celebrates / who observes
<i>Dīpāvalī</i> / <i>Dīwālī</i>	<i>Kārtika</i> amāvasyā	Festival of lights
<i>Mahā-śivarātri</i>	<i>Phālguna</i> (or <i>Māgha</i>) kṛṣṇa caturdaśī	Great night of Śiva
<i>Holī</i>	<i>Phālguna</i> pūrṇimā	Festival of colours

Every festival here has a precise tithi-and-month coordinate. That's the whole point of the *pañcāṅga* system.

5. **The intercalation puzzle — *adhika māsa*.** 12 lunar months $\times$ 29.53 days = 354.36 days. The solar year is 365.24 days. So the lunar calendar drifts behind by ~11 days a year. After 3 years that's 33 days — about a whole month. To re-align, the Indian system inserts an *adhika māsa* ("extra month") roughly every 32.5 months. This is what makes the Indian calendar **lunisolar** rather than purely lunar. (More on this in the cross-cultural lesson, Day 9.)

Activity (15 min) — Festival mapping + formative quiz

Part A (8 min) — pair work. Each pair gets the printed festival list (about 12 festivals). Their job:

1. Mark which lunar month each festival falls in.
2. Identify the tithi and pakṣa.
3. Find ONE festival that no one in the pair had heard of before. Write its month + pakṣa + tithi.

Part B (7 min) — formative quiz, see quizzes/formative.md Part B. Collect and mark within 24 hours.

Wrap-up (5 min)

- One student names a festival their family celebrates; the class names the tithi+pakṣa+month. Two rounds.
- **Project decision deadline:** by the end of today, every student has a project partner (or has confirmed individual work) and a tentative topic. Sticky note on the board.

Student-facing content

There are 12 lunar months. Each is named after the *nakṣatra* in which its *pūrṇimā* falls.

Caitra, *Vaiśākha*, *Jyeṣṭha*, *Āṣāḍha*, *Śrāvaṇa*, *Bhādrapada*, *Āśvina*, *Kārtika*, *Mārgaśīrṣa*, *Pauṣa*, *Māgha*, *Phālguna*.

Two conventions: months end at *amāvasyā* (*amānta*) or at *pūrṇimā* (*pūrṇimānta*). Different regions, different conventions — same astronomy.

Every Indian festival sits at a precise (month + pakṣa + tithi) coordinate. The *pañcāṅga* tells you exactly when.

Homework

- Continue the Moon diary.
- Submit your **project plan paragraph** — 5–8 sentences stating which lunar phenomenon you’ll cover (phases / eclipses / intercalary month / something else), in what format, and what your audience walks away with.

Differentiation

- **One tier up:** *Adhika māsa* deep-dive. Look up the rule: an *adhika māsa* is a lunar month in which no *saṁkrānti* (Sun’s entry into a new zodiac sign) occurs. Why does the rule produce ~7 extra months in 19 years?
- **One tier down:** Memorise only the first 6 month names this week. The rest can come during the project.

Teacher notes

- The *amānta* / *pūrṇimānta* distinction confuses everyone the first time. It’s OK to leave students slightly fuzzy on it — the key insight is that conventions vary regionally, and “the calendar” isn’t one fixed thing.
- Some students will know a festival under a regional name (e.g. *Gudi Pādvā* vs. *Ugādi* — both *Caitra śukla pratipadā*). Honour both.
- The “12 months are named after nakṣatras” point doesn’t need full nakṣatra coverage today — that comes in a later module. Today we just say *what* the naming convention is.

Citation(s) used in this lesson

- General reference: *pañcāṅga* sources (named in class). Paraphrase from *Sūrya-siddhānta*, chapter 14 (computation of *adhika māsa*) — cited by text and chapter name only.